

Unit 3: Cognition

Thought, Reasoning, Decision-making, Problem solving

Cognition

Cognition: The mental activities associated with thought, decision making, language, and other higher mental processes.

Our cognitive activities are subject to many forms of error and distortion.

Basic elements of thought: concepts



A **category** can be defined as a class of similar things (objects or entities) that share one of two things: either an essential core (example: why all science courses are considered “science”) or some similarity in perceptual, biological, or functional properties.

Mental representations we form of categories are called **concepts**.

They allow us to represent a great deal of information about diverse objects, events, or ideas in a **highly efficient manner: Cognitive Economy**.

Concepts **simplify the complex world** of our experience which is full of diversities and wide ranging variations.

Concepts are at the core of intelligent behavior.

Concepts help us establish order in our knowledge base.

Concepts allow us to categorize. **Categorization also allows us to make predictions and act accordingly.**

Also serves **communication** of ideas.

“Without concepts, mental life would be chaotic. If we perceived each entity as unique, we would be overwhelmed by the sheer diversity of what we experience and unable to remember more than a minute fraction of what we encounter. And if each individual entity needed a distinct name, our language would be staggeringly complex and communication virtually impossible. Fortunately, though, we do not perceive, remember, and talk about each object and event as unique, but rather as an instance of a class or concept that we already know something about.”

Smith and Medin (1981)

How are concepts represented?

“What information do we have when we have a particular concept?”

The Classical View

Assumes that categories are well-defined.

All examples or instances of a concept share fundamental characteristics, or **features**.

The classical view of concepts holds that the features represented are **individually necessary** and **collectively sufficient**.

Implications of the classical view:

It assumes that concepts mentally represent lists of features.

Membership in a category is clear-cut: Either something has all the necessary and sufficient features or not.

All members within a category are created equal: There is no such thing as a “better” or “worse” triangle.

Fuzzy Categories

Borderline items

Experiments showed that the psychological assumptions of well-defined categories were not correct.

People disagreed with each other and with themselves (across time) on how to categorise certain items.

Typicality

Even among items that clearly are in a category, some seem to be “better” members than others.

Among birds, for example, crows and sparrows are very typical. In contrast, ostriches and penguins are very atypical.

Borderline members and typicality are two sides of the same coin.

Think of the most typical category member: This is often called the category **prototype**.

Items that are less and less similar to the prototype become less and less typical.

At some point, these less typical items become so atypical that you start to doubt whether they are in the category at all.

Influences of Typicality on Cognition

- **Typical items are judged category members more often** (Hampton, 1979).
- **Speed of categorization is faster for typical items** (Rips, Shoben, & Smith, 1973).
- **Typical members are learned before atypical ones** (Rosch & Mervis, 1975).
- **Learning a category is easier if typical examples are provided** (Mervis & Pani, 1980).
- **In language comprehension, references to typical members are understood more easily** (Garrod & Sanford, 1977).
- **In language production, people tend to say typical items before atypical ones (e.g., “apples and lemons” rather than “lemons and apples”)** (Onishi, Murphy, & Bock, 2008).

Finally, even when given specific instructions to do so, most people cannot generate lists of features that are individually necessary and collectively sufficient to specify membership in a category.

Sources of typicality

Frequency? Not quite.

Family resemblance theory

Items are likely to be typical if they:

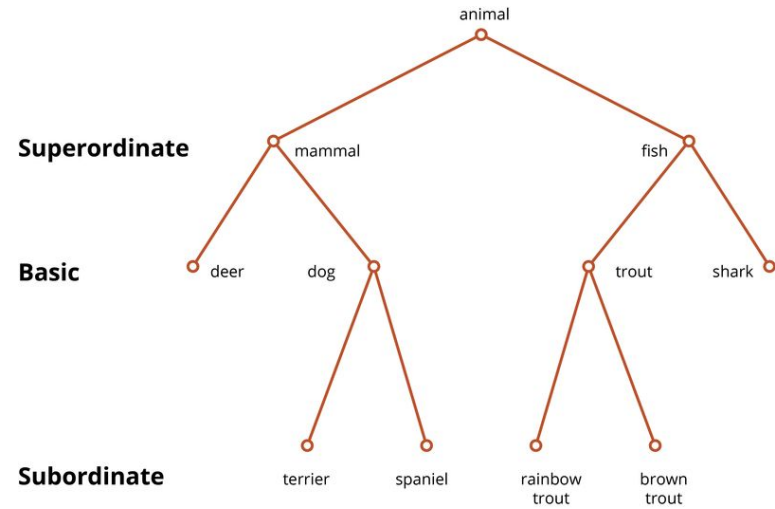
- (a) have the features that are frequent in the category and
- (b) do not have features frequent in other categories.

Concept hierarchies

Brown (1958), a child language researcher, was perhaps the first to note that there seems to be a preference for which category we use to label things.

This preference is referred to as a preference for the basic level of categorization.

The basic level represents a kind of Goldilocks effect, in which the category used for something is not too small (northern brown bear) and not too big (animal), but is just right (bear).



One explanation of the basic-level preference is that **basic-level categories are more differentiated**: The category members are similar to one another, but they are different from members of other categories.

But which level is basic is not universal.

Experts in a domain often have a preferred level that is more specific than that of non-experts.

Birdwatchers see sparrows rather than just birds, and carpenters see roofing hammers rather than just hammers.

The preferred level is not (only) based on how different categories are in the world, but that people's knowledge and interest in the categories has an important effect.

How are concepts represented?

Prototype Theory

Prototype theory suggests that people have a **summary representation of the category**, a mental description that is meant to apply to the category as a whole.

When you try to classify an item, you see how well it matches that weighted list of features.

For example, if you saw something with wings and feathers fly onto your front lawn and eat a worm, you could (unconsciously) consult your concepts and see which ones contained the features you observed. This example possesses many of the highly weighted bird features, and so it should be easy to identify as a bird.

Typical category members have more, higher-weighted features. Therefore, it is easier to match them to your conceptual representation.

Less typical items have fewer or lower-weighted features (and they may have features of other concepts). Therefore, they don't match your representation as well. This makes people less certain in classifying such items.

Borderline items may have features in common with multiple categories or not be very close to any of them.

Exemplar theory

The classical and the prototype views of concepts both hold that concepts are some sort of mental abstraction or summary.

In other words, individual instances are not specifically stored or mentally represented but instead are averaged into some sort of composite representation.

The exemplar view of concepts makes just the opposite assumption: It asserts that concepts include representations of at least some actual individual instances.

The exemplar approach assumes that people categorize new instances by comparing them to representations of previously stored instances, called **exemplars**.

- When you see an object, you (unconsciously) compare it to the exemplars in your memory, and you judge how similar it is to exemplars in different categories.
- For example, if you see some object on your plate and want to identify it, it will probably activate memories of vegetables, meats, fruit, and so on.
- In order to categorize this object, you calculate how similar it is to each exemplar in your memory.
- These similarity scores are added up for each category.
- Perhaps the object is very similar to a large number of vegetable exemplars, moderately similar to a few fruit, and only minimally similar to some exemplars of meat you remember.
- These similarity scores are compared, and the category with the highest score is chosen.

Donker	Blegdav
has thick windows	has steel windows
is red	is purple
divers live there	farmers live there
is under water	is in the desert
get there by submarine	get there by submarine
has fish as pets	has polar bears as pets

The Knowledge-Based View

A person classifying objects and events doesn't just compare features or physical aspects of the objects and events to features or aspects of stored representations.

Instead, the person uses his or her knowledge of how the concept is organized to justify the classification and to explain why certain instances happen to go together in the same category.

When we learn new concepts, we try to connect them to the knowledge we already have about the world.

If you were to learn about a new animal that doesn't seem to eat or reproduce, you would be very puzzled and think that you must have gotten something wrong.

The prototype and exemplar theories don't predict this. They simply say that you learn descriptions or exemplars, and they don't put any constraints on what those descriptions or exemplars are.

However, the knowledge approach to concepts emphasizes that concepts are meant to tell us about real things in the world, and so our knowledge of the world is used in learning and thinking about concepts.

Decision Making, Reasoning

Bounded rationality

Human beings try to make rational decisions (such as weighing the costs and benefits of a choice) but our **cognitive limitations prevent us from being fully rational**.

Time and cost constraints limit the quantity and quality of the information that is available to us.

We only retain a relatively small amount of information in our usable memory.

Limitations on intelligence and perceptions constrain the ability of even very bright decision makers to accurately make the best choice based on the information that is available.

Rational decision?

- (1) define the problem
- (2) identify the criteria necessary to judge the multiple options
- (3) weight the criteria (rank them in terms of importance to you)
- (4) generate alternatives
- (5) rate each alternative on each criterion
- (6) compute the optimal decision.

Consider the letter L. In the English language, is this letter more likely to appear in the first position of a word or the third position of a word? Give your intuition or “gut reaction.”

Availability heuristic

Tversky and Kahneman (1973) argued that when faced with the task of estimating probability, frequency, or numerosity, people rely on **shortcuts or rules of thumb**, known as **heuristics**, to help make these judgments easier.

Availability heuristic: Instances that are more easily thought of, remembered, or computed stand out more in one's mind.

Those instances are particularly salient and hence are deemed to be more frequent or probable.

Examples:

- Our own efforts and behaviors are more apparent and available to us than are the efforts and behaviors of others.
- Risk of flying vs driving.
- Violent crime frequency.



Framing effects

You're driving down the road, and your bike is running low on petrol. You see two petrol pumps nearby, both advertising fuel.

- Pump A advertises a price of ₹100 per liter, with a sign that says, “₹5/liter discount for UPI payments!”
- Pump B advertises a price of ₹95 per liter, with a sign that says, “₹5/liter extra for cash payments.”

Assuming all other factors are equal (e.g., cleanliness of the pumps, fuel quality, waiting times, or your loyalty to a particular brand), which petrol pump would you choose?

Framing effects: People evaluate outcomes as changes from a reference point, their current state. Depending on how their current state is described, they perceive certain outcomes as gains or losses.

We treat losses more seriously than we treat gains of an equivalent amount (whether of money or of some other measure of satisfaction). (**Loss aversion**)

Simply changing the description of a situation can lead us to adopt different reference points and therefore to see the same outcome as a gain in one situation and a loss in the other.

That in turn may lead us to change our decision, not because anything in the problem has changed but simply because the way we describe the situation to ourselves has changed.

Gym membership ad:

"Join now and get 3 months free!"

"Don't miss out on 3 months of free access! Join now"

Which is more persuasive?

Gain Frame: "Switching to renewable energy saves ₹1,000 on your electricity bill annually."

Loss Frame: "Not switching to renewable energy costs you ₹1,000 more on your electricity bill annually."

Anchoring

We know that executive fraud occurs and that it has been associated with many recent financial scandals. And, we know that many cases of management fraud go undetected even when annual audits are performed.

Do you think that the incidence of significant executive-level management fraud is more than 10 in 1,000 firms (that is, 1 percent) audited by major accounting firms?

What is your estimate of the number of Big Four clients per 1,000 that have significant executive-level management fraud? (Fill in the blank below with the appropriate number.)

_____ in 1,000 Big Four clients have significant executive-level management fraud.

The initial starting point will have a huge effect on the final estimates, showing evidence of the phenomenon known as **anchoring**.



Sunk cost effects

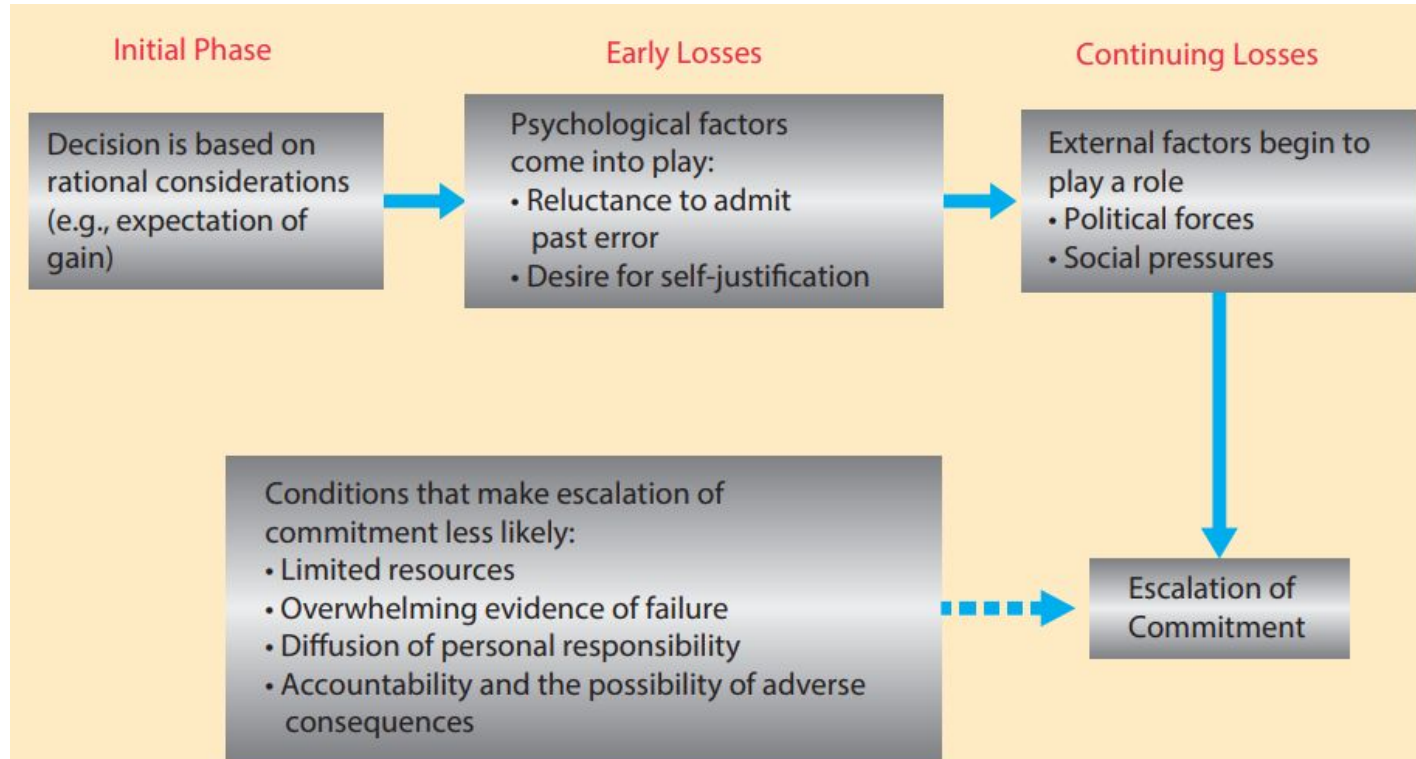
Sunk cost effect: greater tendency to continue an endeavor once an investment in money, effort, or time has been made.

Money spent is already gone.

Whether or not a great deal of money (or time, or energy, or emotion) has been spent does not affect the likelihood of future success.

Those resources have been used, regardless of which option is chosen.

All that should affect a decision, therefore, are the expected future benefits and costs of each option



Hindsight bias

Tendency to consistently exaggerate what could have been anticipated in foresight when looking back (in hindsight) on an event.

Once you know how a decision has turned out, you look back on the events leading up to the outcome as being more inevitable than they really were.



Confirmation bias

Confirmation Bias - searching for positive evidence: The tendency to pay attention primarily to information that confirms existing views or beliefs.

